17:12

An amount of 19 g of molten $SnCl_2$ is electrolysed for some time. Inert electrodes are used. 1.19 g of tin is deposited at the cathode. No substance is lost during electrolysis. If the ratio of the masses of $SnCl_2$ and $SnCl_4$ after electrolysis is x : 261, the value of x is (Sn = 119)

Ans : 1520 g

$$\frac{W_{SnCl_2}}{W_{SnCl_4}} = \frac{x}{261} \Rightarrow x = ?$$

$$(n_{SnCl_2})_{initial} = \frac{19}{(119+71)} = \frac{19}{190} = 0.1\text{ mol.}$$

$$Sn^{2+} + 2e^- \longrightarrow Sn \text{ (cathode)}$$

$0.1 - 0.01 = 0.09$ mol. $\quad$ 0.02F $\quad \frac{1.19}{119} = 0.01$ mol.

$$Sn^{2+} \longrightarrow Sn^{4+} + 2e^- \text{ (Anode)}$$

$0.09 - 0.01$ mol $= 0.08$ $\quad$ 0.01 mol. $\quad$ 0.02F

$n(SnCl_2)_{now} = 0.08$ mol

$n(SnCl_4)_{now} = 0.01$ mol.

Ans.

$$\frac{x}{\cancel{261}} = \frac{0.08 \times 190}{0.01 \times (119 + 2\times 71)} = \frac{1520}{\cancel{261}}$$

$x = 1520$

$$\begin{array}{r} 119 \\ 142 \\ \hline 261 \end{array}$$

17:16

Molar conductivity of aqueous solution of HA is 200 S cm^2 mol^{-1}, pH of this solution is 4. Calculate the value of pK_a(HA) at 25°C.

Given : $\Lambda^{\infty}_{M}(NaA) = 100\,S\,cm^2mol^{-1}$; $\Lambda^{\infty}_{M}(HCl) = 425\,S\,cm^2mol^{-1}$;

$\Lambda^{\infty}_{M}(NaCl) = 125\,S\,cm^2mol^{-1}$

Ans : $pK_a = 4$

Sol^n : $\Lambda_{m,HA} = 200\ S\,cm^2 mol^{-1}$, $pH = 4 \Rightarrow [H^+] = 10^{-4} M = c\alpha$

$$\underset{c(1-\alpha)}{HA_{(aq)}} \rightleftharpoons \underset{c\alpha}{H^+_{(aq)}} + \underset{c\alpha}{A^-_{(aq)}}, \quad K_a = \frac{c\alpha^2}{1-\alpha} = \frac{[H^+]\times\alpha}{1-\alpha} \quad \text{—— ①}$$

Using Kohlrausch law,

$$\Lambda^{\circ}_{m,HA} = (425 + 100 - 125)$$
$$= 400\ S\,cm^2 mol^{-1}.$$

$$\alpha = \frac{\Lambda_m}{\Lambda^{\circ}_m} = \frac{200}{400} = 0.5$$

$$K_a = \frac{10^{-4}\times 0.5}{1-0.5} = 10^{-4}.$$

$pK_a = 4$

Ans

17:22

Imp

A solution containing 1M $XSO_4(aq)$ and 1M $YSO_4(aq)$ is electrolysed. If conc. of X^{2+} is 10^{-z} M when deposition of Y^{+2} and X^{2+} starts simultaneously. Calculate the value of Z.

Given : $\dfrac{2.303RT}{F} = 0.06$

$E^o_{X^{2+}|X} = -0.12V; E^o_{Y^{2}|Y} = -0.24\,V$

Ans: $Z = 4$ Ans

Sol^n:

$\underset{1M}{X^{2+}} + 2e^- \longrightarrow X_{(s)}$

$\underset{1M.}{Y^{2+}} + 2e^- \longrightarrow Y_{(s)}$

X & Y both deposit simultaneously when half cell potential of both become equal.

$E_{X^{2+}/X} = E_{Y^{2+}/Y} \longrightarrow$ just started ppt^n.

$E^o_{X^{2+}/X} - \dfrac{0.06}{2}\log\dfrac{1}{[X^{2+}]} = E^o_{Y^{2+}/Y} - \cancel{\dfrac{0.06}{2}\log\dfrac{1}{1}}$

$-0.12 - 0.03\log\dfrac{1}{[X^{2+}]} = -0.24$

$0.03\log\dfrac{1}{[X^{2+}]} = 0.12$

$\log\dfrac{1}{[X^{2+}]} = 4$

$\Rightarrow [X^{2-}] = 10^{-4}\,M = 10^{-Z}$

$\Rightarrow Z = 4$

During the purification of copper by electrolysis :

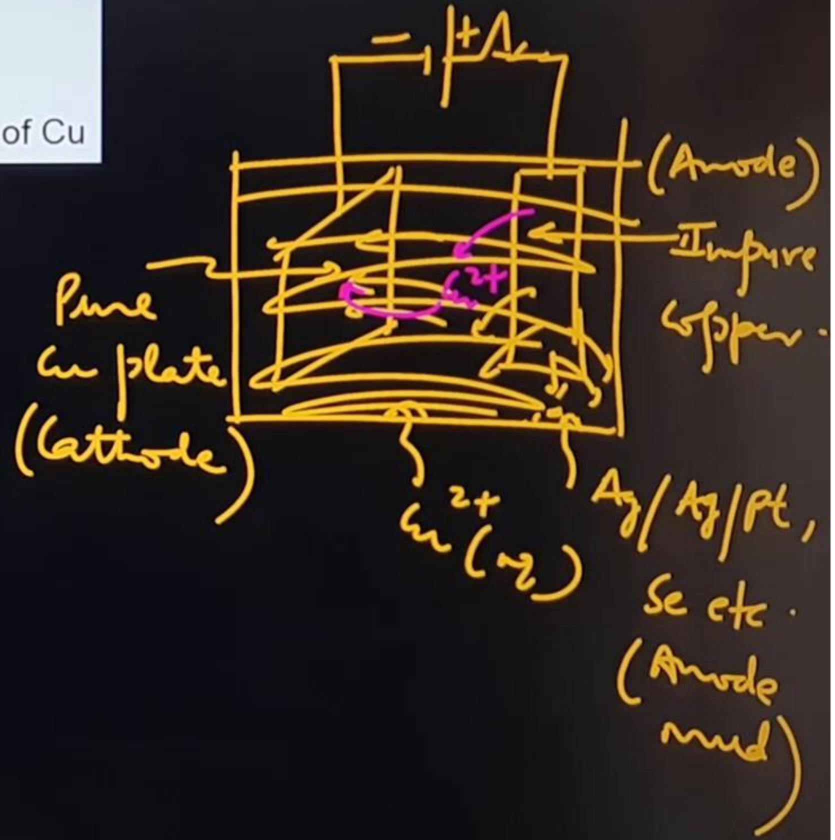
(A) the anode used is made of impure copper ~~ore~~

(B) pure copper is deposited on the cathode

(C) the impurities such as Ag, Au present in solution as ions

(D) concentration of $CuSO_4$ solution remains contant during dissolution of Cu

Sol^n : Ans : A, B, D



17:30

Oxygen and hydrogen gas are produced at the anode and cathode respectively during the electrolysis of fairly concentrate aqueous solution of :

(A) K_2SO_4 (B) $AgNO_3$ (C) H_2SO_4 (D) NaOH

Ans: A, C, D

(A) $K_2SO_4(aq)$: $2H_2O + 2e^- \rightarrow H_2\uparrow + 2OH^-$ (Cathode)

$2H_2O \rightarrow O_2\uparrow + 4H^+ + 4e^-$ (Anode)

(B) $Ag^+ + e^- \rightarrow Ag(s)$ (No H_2 gas)

(C) $2H^+ + 2e^- \rightarrow H_2\uparrow$

$2H_2O \rightarrow O_2\uparrow + 4H^+ + 4e^-$

(D) $2H_2O_{(l)} \xrightarrow{+2e^-} H_2 + 2OH^-$

v.f of $H_2 = 2$

$2OH^- \rightarrow \frac{1}{2}O_2 + H_2O + 2e^-$

v.f. of $O_2 = 4$

17:37

Imp

During electrolysis of $H_2SO_4(aq)$ with high charge density, $H_2S_2O_8$ formed as by product. In such electrolysis 22.4 L $H_2(g)$ and 8.4 L $O_2(g)$ liberated at 1 atm and 273 K at electrode. The moles of $H_2S_2O_8$ formed is :

Ans : 0.25 mol

(A) 0.25 (B) 0.50 (C) 0.75 (D) 1.00

Sol^n:

$H_2(g) \longrightarrow 22.4$ lit

$O_2(g) \longrightarrow 8.4$ L

$H_2S_2O_8 \longrightarrow$ x mol

(Marshall's acid)

1 atm & 273K → STP

$(V_m)_{gas, STP} = 22.4$ lit/mol

$2H^+ + 2e^- \longrightarrow \underline{H_2}\uparrow$ (Cathode)

$2H_2O \longrightarrow \underline{O_2}\uparrow + 4H^+ + 4e^-$ (Anode)

$2H_2SO_4 \longrightarrow 2H^+ + \underline{H_2S_2O_8} + \underline{2e^-}$ (Anode)

No. of 'F' going to cathode = No. of 'F' going to anode.

Eq. of H_2 = Eq. of O_2 + Eq. of $H_2S_2O_8$.

$$\frac{\cancel{22.4}}{\cancel{22.4}} \times 2 = \frac{8.4}{22.4} \times 4 + x \times 2$$

$$2x = 2 - \frac{2.1 \times 4}{5.6} = 2 - \frac{\cancel{2}^{1.5} \times \cancel{4}}{\cancel{8}} = 0.5$$

$x = 0.25$ mol

17:42

$Zn(s) \mid Zn(CN)_4^{2-}(0.5M), CN^-(0.01) \parallel Cu(NH_3)_4^{2+}(0.5M), NH_3(1M) \mid Cu(s)$

Given : K_f of $Zn(CN)_4^{2-} = 10^{16}$, K_f of $Cu(NH_3)_4^{2+} = 10^{12}$,

$|E^o_{Zn|Zn^{+2}} = 0.76V$; $|E^o_{Cu^{+2}|Cu} = 0.34V$, $\dfrac{2.303RT}{F} = 0.06$

The emf of above cell is :

(A) 1.22 V (B) 1.10 V (C) 0.98 V (D) None of these

Ans : 0.98 V

Sol^n : $Zn / Zn^{2+}_{(aq)} \parallel Cu^{2+}_{(aq)} / Cu$

$Zn \longrightarrow Zn^{2+} + \cancel{2e^-}$, $E^o = 0.76V$

$Cu^{2+} + \cancel{2e^-} \longrightarrow Cu$, $E^o = 0.34V$

$Zn + Cu^{2+} \longrightarrow Zn^{2+} + Cu$, $E^o_{cell} = 1.1V.$

Cell rex. of Daniel Cell.

Mernst eq^n,

$E_{cell} = 1.1 - \dfrac{0.06}{2} \log \dfrac{\cancel{5} \times 10^{-9}}{\cancel{5} \times 10^{-13}}$

$= 1.1 - 0.03 \times 4 = 1.1 - 0.12 = 0.98V$

$Zn^{2+} + 4CN^- \longrightarrow [Zn(CN)_4]^{2-}$,

$K_f = 10^{16} = \dfrac{0.5}{[Zn^{2+}] \times (10^{-2})^4}$

$\Rightarrow [Zn^{2+}] = \dfrac{0.5}{10^{16} \times 10^{-8}}$

$= 0.5 \times 10^{-8}$

$= 5 \times 10^{-9} M$

$10^{12} = \dfrac{0.5}{[Cu^{2+}] \times (1)^4}$

$\Rightarrow [Cu^{2+}] = 0.5 \times 10^{-12} = 5 \times 10^{-13} M$

17:53

The conductivity of 0.001 M Na_2SO_4 solution is 2.6×10^{-4} Scm^{-1} and increases to 7.0×10^{-4} Scm^{-1}, when the solution is saturated with $CaSO_4$ (s). The molar conductivities of Na^+ and Ca^{2+} are 50 and 120 S cm^2mol^{-1}, respectivley. Neglect conductivity of used water. What is the solubility product of $CaSO_4$?

(A) 4×10^{-6} (B) 1.57×10^{-3} (C) 4×10^{-4} (D) 2.46×10^{-6}

Sol^n: $[Na^+] = 2\times10^{-3}\,M$

$[SO_4^{2-}] = 10^{-3}\,M$

$K_{Na^+} + K_{SO_4^{2-}} = 2.6\times10^{-4}\ S\,cm^{-1}$

$$\frac{50\times2\times10^{-3}}{1000} + \frac{\Lambda^\circ_{m,SO_4^{2-}}\times10^{-3}}{1000} = 2.6\times10^{-4}$$

$10^{-4} + \Lambda^\circ_m \times 10^{-6} = 2.6\times10^{-4}$

$$\Lambda^\circ_{m,SO_4^{2-}} = \frac{1.6\times10^{-4}}{10^{-6}} = 160\ S\,cm^2\,mol^{-1}$$

$$\Lambda^\circ_{m,Na^+} = \frac{K_{Na^+}\times1000}{C_{Na^+}}$$

$CaSO_4(s) \rightleftharpoons Ca^{2+} + SO_4^{2-}(aq)$

$\quad S \qquad\quad S \qquad (S+10^{-3})\,M$

$\left.\begin{array}{l}\Lambda^\circ_{m,Na^+} = 50\\ \Lambda^\circ_{m,Ca^{2+}} = 120\end{array}\right\}\ S\,cm^2\,mol^{-1}$

$K_{CaSO_4} = (7\times10^{-4} - 2.6\times10^{-4})$

$= 4.4\times10^{-4}$

$$\Lambda_{m,CaSO_4} = \Lambda^\circ_{m,CaSO_4} = \frac{4.4\times10^{-4}\times1000}{S}$$

$$(120+160) = \frac{4.4\times10^{-1}}{S}$$

$$S = \frac{4.4\times10^{-1}}{280} = \frac{\cancel{44}^{11}}{\cancel{28}_{7}}\times10^{-3} = 1.57\times10^{-3}\,M$$

$$\boxed{\begin{array}{l}K_{sp} = S\times(S+10^{-3})\\ = 1.57\times10^{-3}\times2.57\times10^{-3}\\ = 4\times10^{-6}\end{array}}$$

17:57

Equivalent conductivity of $BaCl_2$, H_2SO_4 and HCl, are x_1, x_2 and x_3 S $cm^{-1}eq^{-1}$ at infinite dilution. If conductivity of saturated $BaSO_4$ solution is x Scm^{-1}, then K_{sp} of $BaSO_4$ is:

(A) $\frac{500x}{(x_1+x_2-2x_3)}$ (B) $\frac{10^6 x^2}{(x_1+x_2-2x_3)^3}$ (C) $\frac{2.5\times10^5 x^2}{(x_1+x_2-x_3)^2}$ (D) $\frac{0.25x^2}{(x_1+x_2-x_3)^2}$

$$\lambda^\circ_{eq,BaCl_2} = \lambda^\circ_{eq,Ba^{2+}} + \lambda^\circ_{eq,Cl^-}$$

$$\lambda^\circ_{eq,BaSO_4} = (x_1 + x_2 - x_3)\ S\,cm^2\,eq^{-1}.$$

$$= \frac{\Lambda^\circ_{m,BaSO_4}}{(V.f)_{BaSO_4}} = \frac{\Lambda^\circ_{m,BaSO_4}}{2}$$

$$\Lambda_{m,BaSO_4} = \Lambda^\circ_{m,BaSO_4} = 2(x_1+x_2-x_3) . = \frac{K_{BaSO_4}\times1000}{S}$$

$$S = \frac{500\times x}{(x_1+x_2-x_3)} \Rightarrow K_{sp} = S^2 = \frac{25\times10^4\times x^2}{(x_1+x_2-x_3)^2}$$

$$K_{sp}(BaSO_4) = \frac{2.5\times10^5\,x^2}{(x_1+x_2-x_3)^2}$$

Ans

18:02

Calculate the potential of a half cell having reaction : $Ag_2S\ (s) + 2e^- \rightarrow 2Ag\ (s) + S^{2-}\ (aq)$ in a solution buffered at pH = 3 and which is also saturated with 0.1 M $H_2S(aq)$:

Given : Ksp of $Ag_2S = 10^{-49}$, $K_{a_1} . K_{a_2}$ of $H_2S = 10^{-21}$, $E^\circ_{Ag^+/Ag} = 0.8V$

(A) 1.18 (B) 0.19 (C) -0.19 V (D) none of these

$H_2S \rightleftharpoons 2H^+ + S^{2-}$

$10^{-21} = \frac{(10^{-3})^2 \times (S^{2-})}{0.1}$

$(S^{2-}) = 10^{-16}M$

$\underline{Sol^n}$: $Ag_2S(s) + 2e^- \longrightarrow 2Ag(s) + S^{2-}_{(aq)}$

$\Downarrow$

Ag^+/Ag

$\longrightarrow \underbrace{S^{2-}_{(aq)} / Ag_2S(s) / Ag}$

metal-metal insoluble salt anion half cell.

$E_{S^{2-}/Ag_2S/Ag} = E_{Ag^+/Ag}$ where $(Ag^+)^2 \times (S^{2-}) = Ksp(Ag_2S)$

$(Ag^+) = \left[\frac{10^{-49}}{(S^{2-})}\right]^{1/2}$

$= (10^{-33})^{1/2}$

$= 0.8 - \frac{0.06}{1}\log\frac{1}{(Ag^+)}$

$= 0.8 - \frac{0.06}{2}\log 10^{33}$

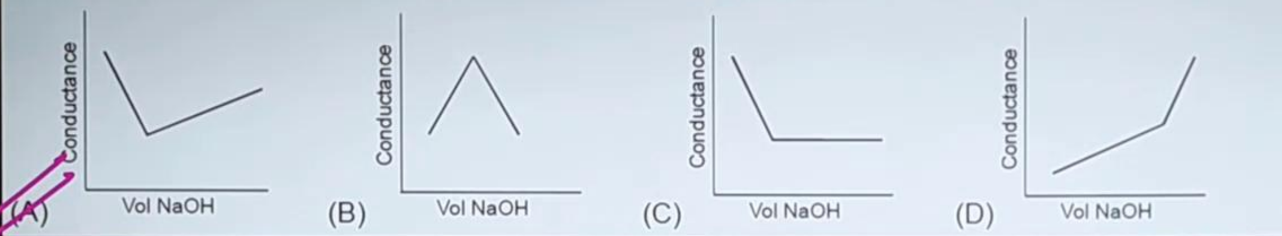
$= 0.8 - 0.03 \times 33$

$= 0.8 - .99$

$= \underline{-0.19V}$

18:07

$HNO_3(aq)$ is titrated with NaOH(aq) conductometrically, graphical representation of the titration as :



Sol^n:

titrant (added drop by drop)

$HNO_{3(aq)} + NaOH \longrightarrow NaNO_{3(aq)} + H_2O$.